ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A4 · ECOSYSTEMS

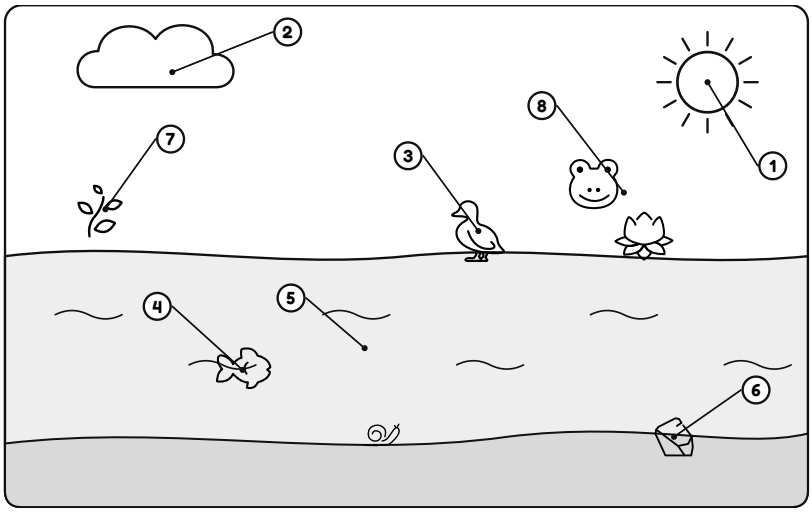
Parts of an ecosystem

★☆☆
Grade 4

Learning goals

- Sort living and non-living parts.
- Order the levels of life.

1 **Look at the pond. Write the name of each numbered part and say if it is living or non-living.** [8]



Number	Name	Living or non-living?
1	Sun (light and heat)	non-living
2	air and clouds	non-living
3	duck	living
4	fish	living
5	water	non-living
6	rock	non-living
7	water plants	living
8	frog	living

2 Put these levels in order, from smallest to biggest. [4]

7	ecosystem
8	biosphere
3	organ
5	population
1	cell
4	organism
2	tissue
6	community

3 Match each interaction to its type. [4]

Draw a line to match each pair, or write the letter.

a heron eats a frog	→ A	**A** feeding
two trees grow towards the same patch of light	→ C	**B** helping each other
a bee carries pollen while it drinks nectar	→ B	**C** competing
a crab hides between mangrove roots	→ D	**D** shelter

ANSWER KEY

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THINK IT THROUGH · A4 · ECOSYSTEMS

When something changes

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Grade 4

Learning goals

- Explain interactions.
- Predict the effects of change.

1 **Explain how each non-living thing helps the living things in a pond.** [3]

Non-living thing	How it helps
sunlight	plants use it to make food; it warms the water
water	fish live in it and take in oxygen from it; frogs lay eggs in it
mud at the bottom	roots of lotus grow in it; worms and snails live in it

2 **A factory pours hot water into a pond. The water gets much warmer and holds less oxygen. Predict what happens to:** [3]

a) the fish

Some fish die or swim away because there is not enough oxygen.

b) the kingfishers that eat the fish

They have less food, so they leave or have fewer young.

c) the people who fish in the pond

They catch fewer fish, so they have less food and money.

3 **Is a pond a system? Explain using the word parts.** [2]

Yes. Its parts (water, plants, fish, birds, sunlight) work together, and changing one part changes the others.

4 **Rafi found a dry seed. It does not move or grow. Is it living or non-living? Explain.** [2]

It is living (but resting). If it gets water and warmth it will germinate and grow.

ANSWER KEY

INVESTIGATE · A4 · ECOSYSTEMS

Do snails prefer damp or dry places?

★★☆ Grade 4

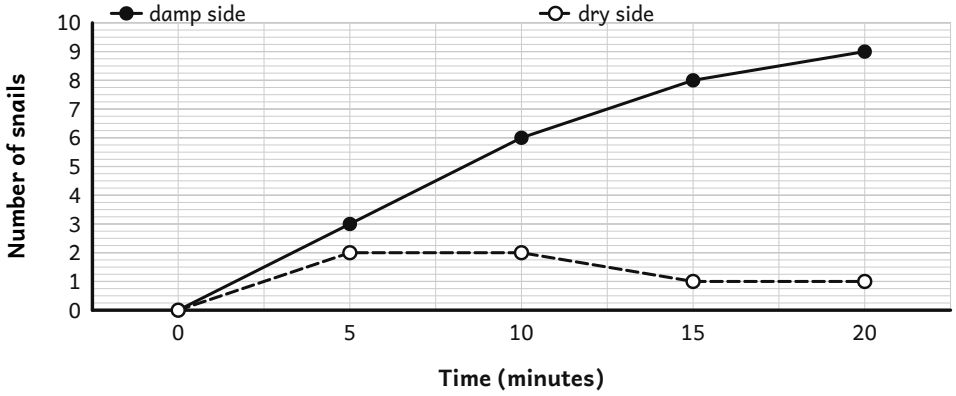
Learning goals

- Use a choice chamber.
- Record results over time.
- Explain results using habitats.

Class 4D put 10 garden snails in the middle of a tray. One half of the tray had damp paper towel; the other half had dry paper towel. Every 5 minutes they counted the snails on each side. Then they put the snails back in the garden.

1 **Here are the results.** [5]

Time (minutes)	0	5	10	15	20
Snails on the damp side	0	3	6	8	9
Snails on the dry side	0	2	2	1	1



a) Plot both lines on the graph.

b) Describe what happened.

More and more snails moved to the damp side; after 20 minutes 9 were on the damp side and only 1 on the dry side.

2 **Why do snails prefer damp places?** [2]

Their soft bodies dry out easily, so they need damp places; they also find food there.

3 **Why did they start with all the snails in the middle?** [1]

So the snails could choose either side fairly.

4 **How could they make the results more reliable?** [1]

Repeat with different snails, or use more snails.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A4 · ECOSYSTEMS

Rice and fish: farming partners

★★★
Grade 4

Learning goals

- Find interactions in a farm ecosystem.
- Use data to compare farms.
- Explain the benefits for people and nature.

Science · Farming · Maths · Economics

In many parts of Bangladesh, farmers keep **fish in their rice fields**. This is called **rice-fish farming**. The field is flooded with water, the rice plants grow, and small fish swim between them.

The rice and the fish help each other. The fish eat insects and weeds that would harm the rice, so the farmer uses less **pesticide**. Fish droppings act as **fertiliser** for the rice. The rice plants give the fish **shade** and shelter. At the end of the season the farmer harvests both rice and fish, which means more food and more money.

Based on research on integrated rice-fish farming in Bangladesh (WorldFish and the Bangladesh Rice Research Institute). Numbers below are example data.

1 Find the interactions in the rice-fish field. [3]

Who helps?	How?
fish help rice	eat insects and weeds; droppings are fertiliser
rice helps fish	gives shade and shelter
water helps both	the fish live in it; the rice roots need it

2 Two farms (example data). [3]

Farm	Rice harvested (kg)	Fish harvested (kg)	Pesticide used (litres)
rice only	500	0	6
rice and fish	520	60	2

a) How much more rice did the rice-fish farm harvest?

20 kg more

b) Why did the rice-fish farm use less pesticide?

The fish ate the insects and weeds.

3 **Why is using less pesticide good for the ecosystem?** [2]

Pesticides can kill helpful insects, fish and birds and pollute the water; less pesticide keeps the ecosystem healthy.

4 **Draw a diagram showing three interactions in a rice-fish field. Use arrows and labels.** [3]

Rice-fish field

e.g. fish → eats insects; fish droppings → feed rice; rice → shades fish; sun → rice.

ANSWER KEY

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REVIEW AND CHALLENGE · A4 · ECOSYSTEMS

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

R	T	B	E	A	H	D	V	H	K	B
C	D	V	P	N	P	H	A	O	Y	A
P	E	A	O	Y	G	H	O	R	L	T
M	A	L	I	E	S	P	Y	G	I	W
S	U	N	L	I	G	H	T	A	V	A
I	C	W	O	S	T	L	L	N	I	T
N	P	S	Y	S	T	E	M	W	N	E
Y	N	R	L	N	B	N	E	O	G	R
W	F	C	O	M	M	U	N	I	T	Y
U	L	O	K	F	Y	S	O	I	L	N
T	W	E	C	O	S	Y	S	T	E	M

cell community ecosystem living organ soil sunlight system water

2 Which is a non-living part of an ecosystem? [1]

- ○ frog
- ● **rock**
- ○ grass
- ○ mushroom

3 All the hilsa fish in the Meghna river are a... [1]

- ○ community
- ● **population**
- ○ biosphere
- ○ organ

4 Draw an ecosystem near your home or school. Label two living and two non-living things, and draw one interaction arrow. [4]

My ecosystem

Any ecosystem, e.g. school garden: tree, bird (living); soil, sunlight (non-living); arrow from bird to tree: nests in it.

5 **Challenge. Explain why scientists say "you cannot change just one thing in an ecosystem".** [2]

Everything is connected. Changing one part (for example, removing trees) changes the food, shelter, light and water for the other parts.